

- 1 (a) Define gravitational potential at a point.

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.....

..... [2]

- (b) A satellite X, of mass M , orbits a planet at a constant distance $4R$ from the centre of the planet, as shown in Fig. 1.1.

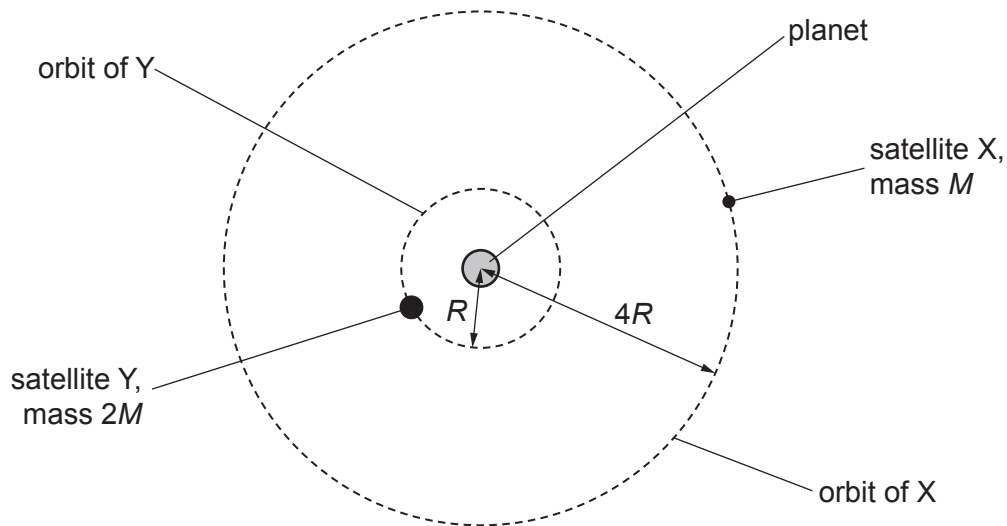


Fig. 1.1 (not to scale)

A second satellite Y, of mass $2M$, orbits the planet with orbital radius R .

The gravitational potential at X due to the planet is $-\Phi$. The planet is a uniform sphere.

- (i) Explain why the gravitational potential at X is negative.

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..... [2]

- (ii) State an expression, in terms of Φ , for the gravitational potential at Y due to the planet.

gravitational potential = [2]

- (iii) Complete Table 1.1 by giving expressions, in terms of some or all of M , R and Φ , for the quantities indicated for each of the satellites X and Y.

Table 1.1

	satellite X	satellite Y
gravitational field strength at satellite due to planet		
gravitational potential energy of satellite		

[4]

[Total: 10]